Alexander Bernstein

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Education

University of California, Santa Barbara Santa Barbara, CA

Ph.D. in Statistics *Sept. 2025*

Dissertation: "Long-Only Minimum Variance Portfolio Composition for Factor Models"

Washington University in St. Louis St. Louis, MO

M.Sc. in Systems Science and Applied Mathematics *Dec. 2016*

B.A. in Mathematics and Economics (Cum Laude) *May 2014*

Technical Skills

Quantitative Methods: Convex & Portfolio Optimization, Factor Models, Asset Pricing (options & bonds), Black-Litterman, Risk Modeling, Time Series Analysis, Stochastic Control, Reinforcement Learning, Asymptotic Theory.

Programming & Tools: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), R, MATLAB, SQL, C, LaTeX, Git, Linux, AWS.

Selected Projects

Factor-Model Portfolio Pipeline: Spec-driven **Python** pipeline for S&P 500 portfolio construction. **PCA** factor models with pluggable covariance estimators (**POET** thresholding, eigenvector shrinkage, GARCH) and return estimators (**Kalman**/EKF filters, robust Student-t, Black-Litterman, L1-adaptive control); constrained mean-variance optimization via **CVXPY** (CLARABEL) with a walk-forward out-of-sample backtester and live rebalancer. **[Add: backtest Sharpe/IR.]**

Quantitative Options Pricing: Implemented **Monte Carlo** (Euler/Milstein) and **finite-difference** (incl. Crank-Nicolson) solvers for the **Heston** and **CEV** models; built **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates for fast pricing and **Greeks** via autograd.

Conditional Covariance Estimation: Built a **Covariance Regression** model for covariate-adjusted covariance of idiosyncratic equity returns — a parsimonious alternative to **MGARCH** that improves stability and cuts parameter count in high dimensions.

Mean Field Game Reinforcement Learning: Implemented a **Q-Learning** solver for the major-minor cyber-security mean field game (Carmona & Delarue, 2018), learning optimal defense policies against an evolving population distribution μ.

Research

Publications:

Banerjee, T., **Bernstein, A.**, Feinstein, Z. (2025). "Dynamic clearing and contagion in financial networks." *European Journal of Operational Research* 321(2), 664–675.

Bernstein, A., Shkolnik, A. (2025). "Asymptotics of Quadratic Forms on a Simplex." *In preparation.*

Selected Talks: INFORMS Annual Meeting (2023); SIAM Conf. on Applied & Computational Discrete Algorithms (2021, poster); CDAR Risk Seminar, UC Berkeley (2020).

Experience

University of California, Santa Barbara Santa Barbara, CA

Teaching Assistant & Graduate Student Mentor *Sept. 2017 – June 2025*

Taught statistical theory, stochastic processes, regression, time series, financial mathematics, and risk theory; mentored undergraduate research in financial mathematics and optimization.

Epic Systems Madison, WI

Technical Services Engineer *Sept. 2014 – Sept. 2015*

Tuned the Epic EHR codebase for large institutional clients; developed fixes for high-availability and regulatory compliance.

Prozess Technologie St. Louis, MO

Intern *May 2013 – May 2014*

Built computational simulations to validate laboratory equipment and calibrated pharmaceutical manufacturing hardware to industry standards.